

CODEX ENGINEERING LAB

VOXEL NATIVE

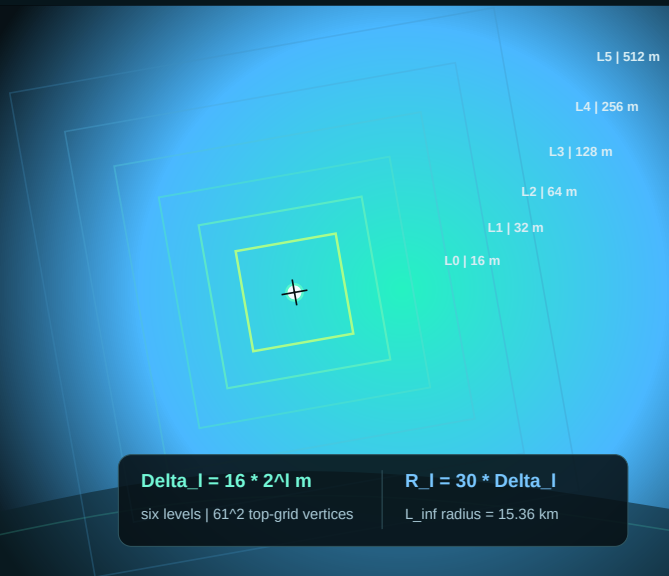
Editable near world. Kilometre-scale horizon.
Fixed budgets. Reproducible evidence.

RUST 2021

BEVY 0.14

WGPU

DETERMINISTIC QA



Project diagram: docs/media/voxel-native-hero.svg.

PROJECT-AUTHORED / FORMULA-FIRST / CONTRACT-BOUND

CODEX ENGINEERING ATLAS

A visual map of the mathematics, representation boundaries, fixed budgets, failure behavior, and evidence discipline behind Voxel Native.

NO RUNTIME RELEASE VERDICT

Runtime gallery pending. This atlas explains current source contracts; it does not convert a generated image, test process, or benchmark into visual acceptance.

INPUT FINGERPRINT / afe0fb87bda7d7a0062cf3df
BUILDER SHA-256 / 8a107d21379a07250229daa1

A technical map, not a cinematic promise

Status words name integration boundaries. They never stand in for a release verdict or a completed visual review.

Label	Meaning	What it does not mean
LIVE	Connected to native runtime and visible or authoritative in its stated profile.	Not automatically accepted across every viewport, seed, route, or profile.
GATED	Implemented and testable behind an explicit mode.	Not default shipping behavior and not proof that its visual gate passed.
PURE LAYER	Compile-registered data implementation with bounded tests.	Not connected to live rendering, physics, edits, or saves.
RESEARCH	A source or candidate studied for an engine question.	Not an implementation claim and not reproduced published results.

CURRENT SOURCE BOUNDARIES

System	Current boundary	Status
Near voxel authority	Full 16^3 chunks own edits, collision, saves, simulation, and exact materials.	LIVE
Planetary far field	One finest parent plus five annuli; six terrain entities; 15.36 km L_inf axis half-extent.	ASTRAL LIVE / NATURAL GATED
Far hydrography	Shared-lattice render-only water and lava; no fluid authority.	GATED
Semantic cohorts	Deterministic sparse L5 silhouettes under an exact selector ceiling.	GATED / VISUAL PENDING
Virtual voxel hierarchy	Fixed-memory middle representation implemented as a pure data layer.	PURE LAYER
Evidence graph	Typed compiler over explicit JSON candidates; native report adapter absent.	TOOLING CONTRACT

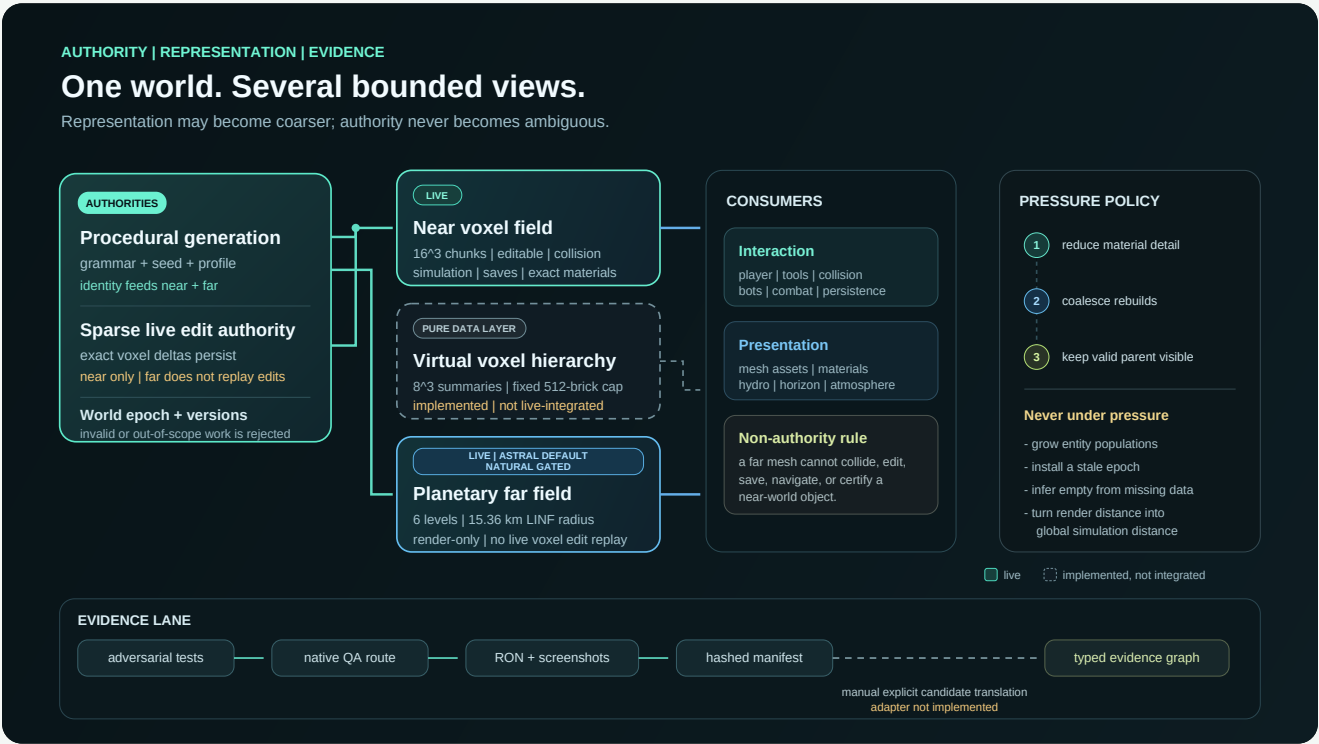
RUNTIME GALLERY PENDING

No live engine screenshot is embedded in this PDF. Four gallery slots remain intentionally empty until reviewed captures are bound to explicit route, seed, viewport, profile, and binary identity.

Source boundary: README.md | docs/CODEX_ENGINEERING_ATLAS.md

One world, several bounded views

Representation becomes coarser with distance. Authority remains explicit: a render proxy cannot silently become edit, collision, or persistence truth.



Geometric reach, fixed representation

The horizon doubles per level while terrain topology, worker ownership, and public generated-payload ceilings remain fixed.

COMPILE-TIME CONTRACT | NOT A BENCHMARK

Geometric reach. Fixed representation.

The horizon doubles per level while population ceilings remain constant.

AXIS HALF-EXTENT / LINF RADIUS | LOG2 SCALE

Level	Axis Half-Extent / Linf Radius (m)
L0	0.48 km
L1	0.96
L2	1.92
L3	3.84
L4	7.68
L5	15.36

$\Delta_l = 16 * 2^l \text{ m}$ | $\|(x,z)\|_{\infty} \leq R_l = 30 * \Delta_l$

NO-CUTOUT TOPOLOGY / HARD CEILING

Bars are drawn to the exact ratios shown; renderer-owned copies are excluded.

Vertices

23,286 / 35,000

Indices

110,760 / 150,000

Generated mesh payload

1,560,768 / 2,280,000 B

$B_{\text{mesh}}(V, I) = 48V + 4I$

position + normal + colour + UV + u32 indices

6 TERRAIN ENTITIES

61 x 61 possible top vertices per level

one asynchronous build in flight

terminal skirt only: 960 vertices + 1,440 indices

Sources: src/planetary_streaming.rs | docs/PLANETARY_STREAMING_PHASE1.md | docs/FAR_TERMINAL_SKIRTS_V1.md

Project diagram: docs/media/planetary-budget-envelope.svg. L_{∞} is the square topology axis half-extent, not a circular Euclidean radius.

SIX-LEVEL RECURRENCE

$$\Delta_l = 16 * 2^l \text{ m}$$
$$R_l = 30 * \Delta_l$$

$l \in \{0,1,2,3,4,5\}$; $R_5 = 15,360 \text{ m}$; exactly six terrain entities.

GENERATED MESH ENVELOPE

$$B_{\text{mesh}}(V, I) = 48V + 4I$$
$$V \leq 35,000; I \leq 150,000$$

Public CPU-generated payload ceiling: 2,280,000 bytes before renderer-owned copies.

EXACT CONSEQUENCE

The no-cutout topology plus terminal L5 skirt produces 23,286 vertices, 110,760 indices, and 1,560,768 generated bytes. More travel changes sample identity, not resident terrain-entity count.

Source boundary: docs/CODEX_ENGINEERING_ATLAS.md | docs/FAR_TERMINAL_SKIRTS_V1.md

04 / INTEGER IDENTITY + LOCAL FLOATS

Seam morphing without planetary float identity

Global sample identity stays in checked signed integers. GPU positions remain local f32 values, and the outer three-cell band converges to the parent lattice.

MORPH COORDINATE

```
d = max(abs(x_local), abs(z_local))
w = 3 * Delta_l
t = clamp((d - (R_l - w)) / w, 0, 1)
```

Chebyshev distance matches the square-ring topology.

DISPLAYED HEIGHT

```
h_display = h_fine
+ s(t) * (bilerp(h_parent) - h_fine)
```

Parent interpolation is evaluated on the next coarser global integer lattice.

SMOOTHSTEP

```
s(t) = t^2 * (3 - 2t)
s(0)=0; s(1)=1
s'(0)=s'(1)=0
```

Endpoint slope is zero; this reduces a sharp grade change but is not a visual verdict.

EUCLIDEAN PARTITION

```
x = 16q + r
q = x div_e 16
r = x mod_e 16; 0 <= r < 16
```

Truncation toward zero is forbidden for negative world identity.

SIGNED COORDINATES ARE ALGEBRA

World x	Euclidean chunk q	Local r	Why it matters
15	0	15	Last local voxel before the positive chunk boundary.
0	0	0	Origin belongs to chunk zero.
-1	-1	15	Negative neighbor, not chunk zero.
-16	-1	0	Exact negative chunk boundary.

FAIL CLOSED

Checked i64/i128 intermediates reject unrepresentable operations instead of wrapping a coordinate into another place. The same rule is shared by chunks, rings, material cells, virtual bricks, and semantic supertiles.

Source boundary: docs/CODEX_ENGINEERING_ATLAS.md | src/chunk.rs

05 / STREAMING OWNERSHIP

Toroidal sampling and a fail-closed Near handoff

Every level moves one fixed source window. Coverage removal is conservative: uncertainty keeps the parent instead of opening a sky hole.

FIXED CPU CENTRE-SAMPLE WINDOW | EXACT PER-RING COUNTS

Travel rotates a 65 x 65 window-not its budget.

Logical overlap is retained in place; only entering samples are replaced.

LIVE CENTRE-SAMPLING CONTRACT

NOT A GPU PARTIAL-UPDATE CLAIM

mesh assembly and upload remain separate work

01 | One-cell axial shift

shift = (+/-1, 0) or (0, +/-1)

65 new 4,160 reused

02 | One-cell diagonal shift

shift = (+/-1, +/-1)

129 new 4,096 reused

03 | Bounded full fallback

either axis |shift| >= 65, or incompatible/non-grid

4,225 refill 0 reused

$N = 65^2 = 4,225$ reused = $(65 - |dx|)(65 - |dz|)$ new = $N - \text{reused}$ valid while $|dx|, |dz| < 65$

Entering strips are enlarged for legibility; the numbers are exact 65 x 65 centre-sample populations per ring.
Sources: src/planetary_streaming.rs | docs/PLANETARY_STREAMING_PHASE1.md | docs/CODEX_ENGINEERING_ATLAS.md

Project diagram: docs/media/toroidal-cache-reuse.svg. Exact centre-sample reuse populations; mesh assembly and GPU upload remain separate work.

NEAR READINESS BYTES

33^2 bools = 1,089 B
ceil(60^2/64) * 8 = 456 B
total = 1,545 B

The finest parent mask is a fixed 3,600-bit workset.

ASYMMETRIC STABILITY

coverage gain: wait 0.5 s
coverage loss: restore immediately

Temporary overlap is preferred to a missing parent cell.

HANDOFF DECISION TABLE

Observed state	Parent decision	Failure behavior
Current request proves stable Near coverage	Remove covered parent cell after 0.5 s.	Promotion is temporally stable.
Coverage unknown or missing	Retain parent.	No speculative hole.
Coverage stale, invalid, or unrepresentable	Retain parent; reject result.	Old or unsafe work cannot become geometry.
Coverage is lost	Restore parent immediately.	Safety recovers faster than cosmetic demotion.

ROLLBACK BOUNDARY

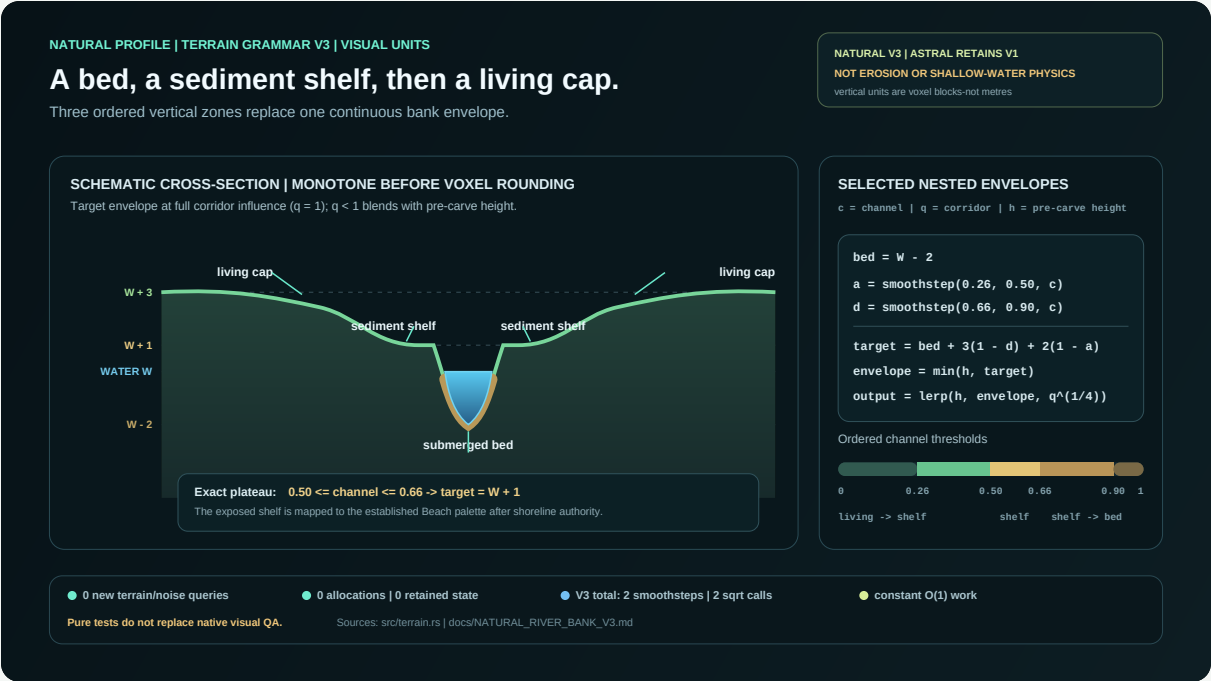
A large incompatible anchor shift refills the same fixed window. It does not allocate a second unbounded history, and existing visible terrain remains until a complete validated replacement can install.

Source boundary: docs/CODEX_ENGINEERING_ATLAS.md | docs/PLANETARY_STREAMING_ARCHITECTURE.md | docs/media/toroidal-cache-reuse.svg

06 / GATED DESCRIPTIVE LAYERS

Hydrography, river-bank grammar, and sparse silhouettes

Both layers are deterministic and budgeted presentation. Neither can become fluid, voxel, edit, collision, navigation, physics, or save authority.



Project diagram: docs/media/river-bank-v3-cross-section.svg. Natural V3 uses authored voxel-block visual units; it is not an erosion or shallow-water simulation.

HYDRO + BANK ENVELOPE

wet ring: $V \leq 3,721$
 $I \leq 21,600$
 $B \leq 265,008$ bytes

GATED descriptive fluid plus Natural V3 bank grammar; Astral retains V1.

COHORT SELECTOR

one selected cell per $8 * 8$ supertile
public candidates $\leq 9 * 9 = 81$
payload $\leq 104,976$ bytes

Euclidean supertile identity is replayable across negative coordinates.

ATOMIC WORKER ENVELOPE

Layer combination	Maximum generated payload	Install rule	Rollback
Terrain + Hydro	653,008 B	Validate complete identity and both CPU payloads before asset creation.	Hydro gate off
Optional L5 cohorts	104,976 B	One combined L5 cohort entity; reject impossible shape or budget excess.	Cohort gate off
Terrain + Hydro + cohorts	757,984 B	Any stale, malformed, or over-budget component publishes none of the new meshes.	Disable gates independently

PROOF BOUNDARY / Unit tests and exact ceilings establish mechanics. Native Natural/Astral screenshots and telemetry must still be inspected together; average FPS cannot promote either layer.

Source boundary: docs/FAR_HYDROGRAPHIC_CONTINUITY_V1.md | docs/NATURAL_RIVER_BANK_V3.md | docs/FAR_SEMANTIC_COHORTS_V1.md | docs/media/river-bank-v3-cross-section.svg

Fixed-memory virtual bricks

The hierarchy explores a conservative layer between exact chunks and the height-only far field. It is implemented, bounded, and deliberately not live-integrated.

SUMMARY PAYLOAD

cell = u16 material + u8 occupancy + u8 error

cell = 4 B

brick = 8^3 * 4 = 2,048 B

Fixed X-contiguous index: i(x,y,z) = x + 8z + 64y.

NATIVE CACHE ACCOUNTING

512 bricks

B_cache = 1,093,632 B on verified 64-bit native

active tickets <= 128; ticket bytes = 7,168 B

wasm32 computes its own compile-time layout values.

REDUCTION

raw / summary = (2^L)^3 = 2^(3L)

L2 = 64x

L4 = 4,096x

Relative to the same four-byte raw voxel/material payload.

CURRENT STATUS

PURE LAYER. Compile-registered and tested, but not connected to renderer, physics, saves, or the live streaming scheduler.

PRESSURE AND AUTHORITY

Invariant	Mechanism	Failure or rollback boundary
Resident population never grows with travel	Fixed 512-slot second-chance clock plus sorted fixed-capacity lookup.	Reconstructible summaries are evicted deterministically.
Sparse edits remain authoritative	Generator summaries and edit overlay are separate ownership domains.	Cache pressure never evicts edit records.
Uncertainty does not become empty space	Positive mass quantizes to at least one; error-only cells stay non-empty.	Request refinement instead of manufacturing a hole.
Arithmetic cannot wrap the budget	Checked byte-accounting returns None on multiplication overflow.	Reject install; do not understate memory.
Rollback remains local	The pure module is reconstructible and has no live pipeline ownership.	Remove integration without rewriting saves or Near authority.

Source boundary: docs/VIRTUAL_VOXEL_HIERARCHY.md | docs/CODEX_ENGINEERING_ATLAS.md

Road-first city planning math

Candidate scoring is bounded, interpretable, and cheap enough to avoid a world scan. Its weights are engineering heuristics, not physical constants or new general theorems.

AUTONOMOUS CONSTRUCTION | LIVE PROJECT-AUTHORED HEURISTIC

Hard gates first. Transparent scoring second.

Hard spatial filters precede a bounded, deterministic weighted comparison.

INTERPRETABLE | TESTABLE

NOT A PHYSICAL MODEL

weights encode project design priorities

HARD PREFILTER

Candidate footprint must be:

inside primary bounds

anchored in loaded world data

outside reserved project footprints

clear of protected road corridors

not blocking city footprint (road)

not duplicating an existing corridor

FAILED GATE -> FILTERED OUT

WEIGHTED SCORE

Valid-site composite

Normalized terms stay visible; the centre term gently prefers compact plans.

site_score =

+ 2.50 flatness

+ 2.40 road_access

+ 1.80 district_balance

+ 1.35 route_fit

+ 0.55 block_fit

+ 4.00 road_anchor_alignment

+ 2.50 semantic_anchor

- 0.0005 centre_distance

road_access = 0.35 * proximity + 0.65 * frontage

alignment = clamp((|dot| - 0.72) / 0.28) * clamp(1 - distance / 24)

Project-kind branches set inapplicable terms to fixed neutral values.

SELECTION

Choose the maximum

The same candidate order and world state replay the result.

max_by

sa.total_cmp(sb)

selected site

bounded candidate set

BASE-HELPER SAFETY SENTINEL

!inside_bounds OR !player_clearance -> -10,000

Candidate generation normally prefilters invalid sites; later composite terms mean -10,000 is not claimed as an exact final composite score.

ACCESS-ROAD ROUTE SUBSCORE

route_fit = clamp(1 - 0.55*clamp(avg_step/5) - 0.30*clamp(max_step/9) - 0.15*clamp(max(range-18,0)/34), 0, 1)

Sources: src/bots.rs | docs/CODEX_ENGINEERING_ATLAS.md | docs/CITY_PLANNER_MATH.md

fewer than 2 heights -> 0.5

Project diagram: docs/media/city-site-score.svg. Six hard filters precede a weighted score and deterministic maximum selection.

ROAD GRADE FIT

route_fit = clamp(1 - 0.55*clamp(avg_step/5,0,1) - 0.30*clamp(max_step/9,0,1) - 0.15*clamp(max(height_range-18,0)/34,0,1), 0,1)

A bounded terrain profile penalizes poor access inside the deterministic site score before voxel work.

SMOOTH RAISED DECK

s(t) = t^2 * (3 - 2t)

deck_y(t) = round(start_y + (end_y - start_y) * s(t))

The same cubic family supplies a cheap deterministic grade envelope.

PLANNING BOUNDARY

Decision	Chosen contract	Rejected failure mode
Roads before buildings	Reserve road corridors, bind frontage, then place structures.	Buildings first and roads carved through them later.
Bounded candidate set	Fixed candidate counts and local road/lot probes.	Full-world voxel scans or per-frame city rebuilds.
Clearance	Hard admission before project creation; execution pauses if clearance becomes unsafe.	Bots constructing around the active player or ship.
Pressure	Edit queues yield when chunk streaming is behind.	Hiding city cost inside horizon backlog.

Source boundary: docs/CITY_PLANNER_MATH.md | src/bots.rs | src/city.rs | docs/media/city-site-score.svg

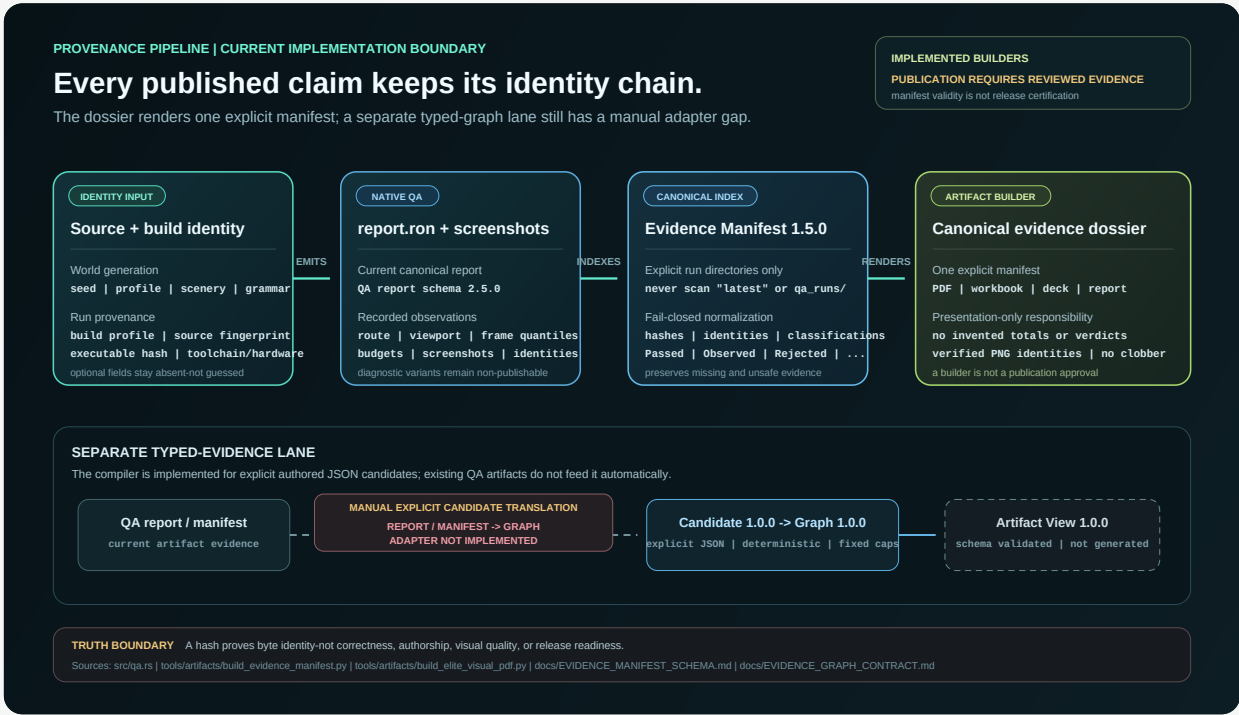
SOURCE-FIRST / BOUNDED / REVERSIBLE

09 / 15

09 / DETERMINISTIC EVIDENCE IDENTITY

Technical truth is separate from presentation

A caller supplies a bounded local alias but cannot choose an authoritative node identity. Canonical source bytes and typed relationships remain visible downstream.



Project diagram: docs/media/evidence-lineage.svg. Source/build identity flows through native QA and canonical manifests; the report-to-graph adapter is still absent.

AUTHORITATIVE NODE IDENTITY

```
node_id = kind : sha256(canonical_json(identity))
```

Canonical JSON uses sorted keys, finite values, deterministic scalar spelling, UTF-8, and no insignificant whitespace. Identity is not correctness or release readiness.

TWO STATUS DIMENSIONS

Evidence classification	Task state	Non-equivalence
Passed / Observed / Rejected / Planned / Blocked	planned / ready / running / blocked / review / complete / cancelled	A complete task may produce a Rejected visual result.
Describes evidence	Describes workflow	Blocked evidence is not task blocked, and missing evidence is not zero.

INTEGRATION LIMIT

The typed graph currently compiles explicit bounded JSON candidates. Native report.ron and QA manifests are not automatically translated; the manifest-to-candidate adapter remains unimplemented.

Source boundary: docs/EVIDENCE_GRAPH_CONTRACT.md | docs/EVIDENCE_MANIFEST_SCHEMA.md | docs/media/evidence-lineage.svg

10 / FIXED BUDGETS + FAILURE CONTAINMENT

Every ambitious system publishes its stop condition

The useful promise is not infinite dense voxels at zero cost. It is bounded work with graceful representation changes and a reversible boundary.

System	Hard envelope	Failure behavior	Rollback boundary
Far terrain	6 levels; V <= 35,000; I <= 150,000; B <= 2,280,000	Reject stale, malformed, over-budget, or identity-invalid build before install.	Disable gated profile route; retain Near and current parent.
Near handoff	1,545 B readiness/mask workset	Unknown or stale coverage retains parent; loss restores immediately.	Refill same fixed window; no history growth.
Hydro	6 fluid entities; 1,590,048 generated B total	Atomic terrain/fluid rejection; dry ring emits no fluid entity.	Hydro gate off; no persisted data migration.
Semantic cohorts	81 candidates; 1 entity; 104,976 B	Fail-closed parsing; impossible shape or overflow rejects payload.	Cohort gate off; terrain and saves unchanged.
Virtual bricks	512 residents; 128 tickets	Deterministic eviction; checked accounting; authority remains separate.	Remove pure integration; reconstruct summaries.
Evidence graph	64 inputs; 12k nodes; 32k edges; 16 MiB	All-or-nothing compile; no permissive legacy mode.	Remove tools/evidence without changing inputs/builders.
Agent control	64 KiB control file; bounded strings/vectors; strict sequence	Reject stale/reused identity, non-finite pose, unsafe path, or oversized input.	Explicit isolated session; runtime exit is separate.
City planner	Fixed candidate counts; local probes; bounded edit queues	Hard-invalid sites do not consume voxel edit budget.	Yield planning when streaming pressure rises.

PRESSURE ORDER

Reduce detail, cadence, or promotion radius before shortening the far silhouette, evicting authority, or allowing queues to grow with travel distance.

UNKNOWN MEANS UNKNOWN

Corrupt, stale, unsupported, or unobserved state remains a visible diagnostic. It is never converted to zero or silently labeled healthy.

NOVEL SOLUTION RULE

Record baseline, alternatives, fixed budget, measured distribution, invalidating assumptions, failure mode, and rollback boundary. Simpler code that wins the real metric is the more advanced solution.

Source boundary: docs/ELITE_WORLD_SYSTEMS_STANDARD.md | docs/PLANETARY_STREAMING_ARCHITECTURE.md | src/agent_control.rs

Research becomes reversible engine work

Each step preserves the distinction between an idea, a bounded implementation, an observation, and an accepted claim.

01	RESEARCH QUESTION	Name the world-system question and the exact user-visible failure.
02	AUTHORITY BOUNDARY	Declare who owns edit, collision, persistence, rendering, and simulation state.
03	FIXED BUDGET	Cap bytes, entities, work, queues, tasks, and per-frame install population.
04	IMPLEMENT	Carry world identity, epoch, sequence, and stale-result rejection through async work.
05	ADVERSARIAL TEST	Exercise negative/extreme coordinates, order independence, pressure, overflow, and malformed input.
06	ONE-BINARY ROUTES	Compare like-for-like arms from the same release executable and explicit route identity.
07	INSPECT TOGETHER	Read full-size screenshots and telemetry together; neither can substitute for the other.
08	RETAIN / REVISE / ROLL BACK	Promote only the narrow claim supported by evidence; preserve rejected observations.

WHY SCREENSHOTS

A beautiful frame can conceal unbounded work, stale authority, or a broken alternate route.

WHY TELEMETRY

Clean counters can conceal a hole, pop, unreadable UI, unstable silhouette, or wrong material.

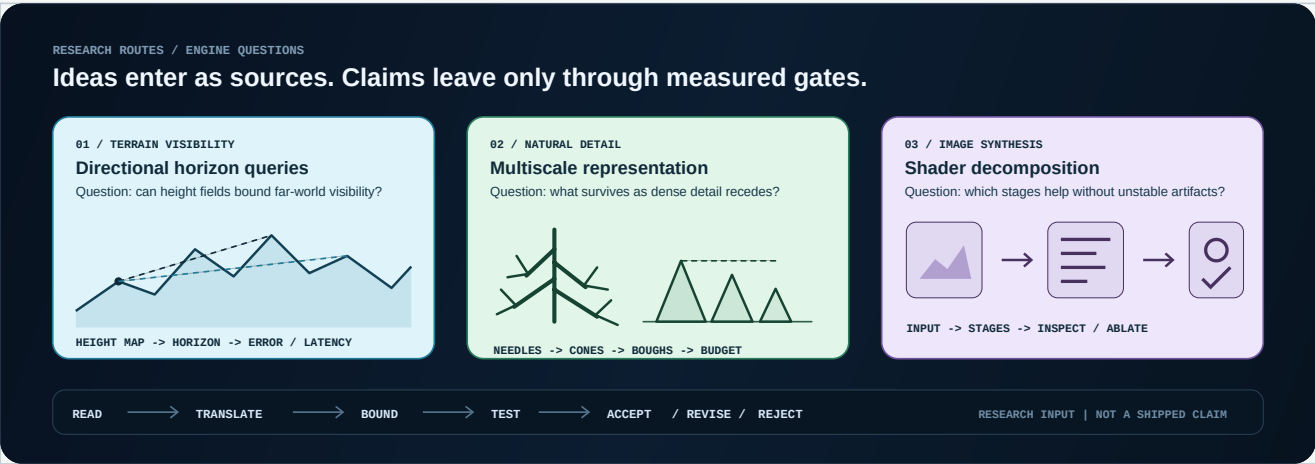
CODEX OPERATOR CONTRACT

- Batch source changes, run deterministic checks, and use one deliberate native rebuild when compiled Rust changes.
- Keep user saves and unrelated dirty files outside the experiment boundary.
- Record exact toolchain, target, profile, seed, route, viewport, binary identity, and known limits.
- Do not label a fallback transport as direct or a completed PNG as a visual pass.

Source boundary: README.md | docs/ELITE_WORLD_SYSTEMS_STANDARD.md | docs/RESPONSIVE_VISUAL_QA.md

Ideas enter as sources; claims leave through gates

Research routes are traceable to original publishers. Studying a technique does not mean Voxel Native ships it or reproduces its published result.



Research input	Engine question	Current claim
Virtual Horizon Method	Can a height-map abstraction bound far-world visibility under an accuracy/latency trade-off?	RESEARCH - no VHM runtime claim.
Multiscale pine rendering	Which representation survives as dense natural detail recedes?	RESEARCH - no multiscale tree-shader runtime claim.
Generative Adversarial Shaders	Which decomposed shader stages help without unstable temporal artifacts?	RESEARCH - no learned shader ships.

TRANSFER CHECK

Stage	Required record	Disallowed shortcut
READ	Original publisher URL, bounded local notes, exact question.	Treating a secondary summary as implementation proof.
TRANSLATE	Authority boundary and engine-specific failure metric.	Copying a paper label onto unrelated code.
BOUND	Work, memory, population, and rollback envelope.	Average cost without a hard ceiling.
TEST	Deterministic/adversarial checks plus same-binary native routes where visual.	Unit tests alone for perceptual acceptance.
PROMOTE	Narrow claim, evidence identity, known limits, and reviewed images.	A link, screenshot, or benchmark promoted by aesthetics.

RESEARCH IS VISIBLE

The Voxel Discovery Atlas tracks adopted, prototype, deferred, and rejected routes across clipmaps, sparse volumes, virtual texturing, splatting, watersheds, natural structures, QA, and implicit geometry.

Source boundary: docs/VOXEL_DISCOVERY_ATLAS.md | docs/FAR_WORLD_RENDERING_RESEARCH.md

13 / DECLARED LIMITS

Runtime gallery pending

The empty gallery is deliberate. This source-authored atlas does not scan QA runs or select a convenient image; visual evidence requires explicit manifest identity and full-size review.

NATURAL OVERVIEW

PENDING - matched same-binary visual gate not attached

ASTRAL OVERVIEW

PENDING - reviewed capture not attached

NEAR / FAR HANDOFF

PENDING - route + telemetry + visual review required

BUILD STUDIO

PENDING - interaction and responsive matrix evidence required

KNOWN LIMITS THAT REMAIN VISIBLE

Boundary	Declared limit	Next proof
Natural far terrain	Explicitly gated; matched visual acceptance remains pending.	Same release binary, fixed routes, required viewport/DPI cells, full-size review.
Virtual hierarchy	Pure data layer; no live renderer, physics, save, or scheduler connection.	Integration contract plus cross-tier edit and identity proof.
Far procedural layers	Far does not replay sparse user edits today.	Bounded edit projection with conservative invalidation.
Hydro	No flow, depth, refraction, buoyancy, waves, foam, or collider.	Separate feature contracts; no inference from colour quads.
Semantic cohorts	Placeholder silhouettes; composition and pop acceptance remain open.	Natural/Astral on/off routes with telemetry agreement.
Evidence graph	No native report/manifest adapter; no production-size benchmark.	Explicit adapter and throughput evidence without semantic drift.
Responsive QA	Several viewport and 100/150/200 percent scale cells remain outstanding.	Capture and inspect every required matrix cell.
License	No reuse license has been declared.	Maintainer-selected license before redistribution rights are implied.

PROMOTION RULE

Name mode and authority; bind source and binary identity; record seed/profile/route/viewport; validate populations and stale-result identity; inspect every image; report distributions; retain rejected evidence; roll back when required.

Source boundary: docs/CODEX_ENGINEERING_ATLAS.md | docs/RESPONSIVE_VISUAL_QA.md

14 / REFERENCES + VERIFICATION

Official routes and final artifact checklist

Links are research and platform references. They are not evidence that an external technique or result ships in Voxel Native.

Rust language Ownership, checked arithmetic, deterministic native systems https://www.rust-lang.org/	Rust Euclidean division Signed chunk, ring, brick, and supertile mapping https://doc.rust-lang.org/std/primitive.i32.html#method.div_euclid
Bevy ECS, application lifecycle, assets, and native rendering integration https://bevyengine.org/	wgpu Portable graphics abstraction used by the native renderer https://wgpu.rs/
GPU Gems 2 - Geometry Clipmaps Primary terrain-LOD research route; not a runtime equivalence claim https://developer.nvidia.com/gpugems/gpugems2/part-i-geometric-complexity/cha-pter-2-terrain-rendering-using-gpu-based-geometry	Virtual Horizon Method - IBPSA Directional horizon-query research input https://publications.ibpsa.org/conference/paper/?id=bs2025_1302
Multiscale pine rendering - Graphics Interface Scale-dependent natural-detail research input https://graphicsinterface.org/proceedings/gi2000/gi2000-19/	Generative Adversarial Shaders - arXiv Shader decomposition and ablation-discipline research input https://arxiv.org/abs/2306.04629
NIST FIPS 180-4 SHA-256 reference for deterministic evidence identity https://csrc.nist.gov/pubs/fips/180-4/upd1/final	

FINAL VERIFICATION CHECKLIST

Owner	State	Check
BUILDER	[x]	All 29 immutable source snapshots bounded, anchored, and fingerprinted.
BUILDER	[x]	All eight passive ASCII SVGs safely parsed and embedded with repo-relative captions.
BUILDER	[x]	Exactly 15 strict A4 pages; envelope, compression, metadata, IDs, and links validated.
BUILDER	[x]	PDF text rejects workstation paths, controls, replacement glyphs, and Unicode dashes.
BUILDER	[x]	NO RUNTIME RELEASE VERDICT and RUNTIME GALLERY PENDING remain extractable text.
OPERATOR	[]	Render every page to PNG with Poppler using the final output bytes.
OPERATOR	[]	Inspect every page at full size for clipping, overlap, black squares, broken SVG text, and weak contrast.
OPERATOR	[]	Verify page numbering, headers/footers, link readability, formula legibility, and empty gallery honesty.
OPERATOR	[]	Record final PDF SHA-256 only after the rendered-page review is complete.

FINAL BOUNDARY

Aggregate source fingerprint: afe0fb87bda7d7a0062cf3df354388b2db462e2b1f7c5a6554199db233f3c0cd. Builder SHA-256: 8a107d21379a07250229daa176ef26c0634f360aed9809a5061c83b1a4b84c8c. The PDF is complete as a project-authored technical atlas only after separate rendered-page inspection; the runtime gallery and runtime release verdict remain pending.

Source boundary: 29 immutable inputs enumerated by the builder contract; authoritative Rust anchors include src/chunk.rs | src/bots.rs | src/city.rs | src/agent_control.rs.